Check for updates

Space Weather

RESEARCH ARTICLE
10.1029/2022SW003202

Improved Model for GIC Calculation in the Mexican Power Grid

R. Caraballo[1], J. A. González-Esparza[2], C. R. Pacheco[3], and P. Corona-Romero[2,4]

[1]Instituto de Geofísica, Unidad Michoacan, Universidad Nacional Autónoma de México, Morelia, Mexico, [2]SCiESMEX, LANCE, Instituto de Geofísica, Unidad Michoacan, Universidad Nacional Autónoma de México, Morelia, México, [3]Gerencia de Ingeniería Especializada, Comisión Federal de Electricidad, Mexico City, Mexico, [4]CONACYT, Instituto de Geofísica, Unidad Michoacan, Universidad Nacional Autónoma de México, Morelia, México

Abstract We present the first observations of geomagnetically induced currents (GIC) in the Mexican power grid and an improved model to calculate them. The new model comprises ca. 250 substations working at various voltage levels, a methodology to estimate geomagnetic disturbances (δB) at different points throughout the Mexican territory, and a 1D piecewise model that considers lateral variations in the ground conductivity. This is an improvement of a former uniform conductivity model presented previously to calculate our first GIC estimates (Caraballo et al., 2020). We compared the observed and calculated GIC between August and November 2021 at a coastal 400 kV substation. During this interval, five geomagnetic storms occurred (G1 and G2). The observed GIC exceeded 10 A during the most strong event; this shows a clear grid response even under weak geomagnetic perturbations that occurred during the solar minimum. Further comparison with the results of the former model suggests that the new 1D piecewise model yields better GIC estimates for the Mexican power grid.

Plain Language Summary We present observations of geomagnetically induced currents (GIC) in the Mexican power grid and an improved model to calculate them. The new model considers multiple operational voltages as well as lateral and vertical variations in ground conductivity. Additionally, we analyzed GIC observations at a given substation from August to November 2021 and the grid response during five geomagnetic storms that occurred during this period. The observed GIC shows a clear grid response even during weak geomagnetic perturbations. Regarding calculated GIC, comparisons with the former model output suggest that the 1D piecewise model can be a valuable tool for providing GIC estimates for the Mexican power grid.

Key Points:

- We compared geomagnetically induced currents (GIC) observations and estimates for the Mexican power grid during 2021 (near solar cycle minimum)
- An improved model to calculate GIC yields better estimates for the Mexican 400/230 kV grid
- Measured GIC at a coastal substation exceeded 10 A during a G2 geomagnetic storm that occurred on November 2021

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

R. Caraballo,
rcaraballo@igeofisica.unam.mx

Citation:

Caraballo, R., González-Esparza, J. A., Pacheco, C. R., & Corona-Romero, P. (2023). Improved model for GIC calculation in the Mexican power grid. *Space Weather*, *21*, e2022SW003202. https://doi.org/10.1029/2022SW003202

Received 21 JUN 2022
Accepted 6 AUG 2023

Author Contributions:

Conceptualization: R. Caraballo
Formal analysis: R. Caraballo
Investigation: R. Caraballo
Methodology: R. Caraballo
Project Administration: J. A. González-Esparza
Software: R. Caraballo
Validation: R. Caraballo
Visualization: R. Caraballo
Writing – original draft: R. Caraballo
Writing – review & editing: R. Caraballo, J. A. González-Esparza, C. R. Pacheco, P. Corona-Romero

© 2023. The Authors.
This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

1. Introduction

Geomagnetically Induced Currents (GIC) might threaten electric power grids at low-to-mid geomagnetic latitudes. In the last 15 years, some studies have appeared in the literature due to efforts to address the calculation of GIC amplitudes in power grids at these geomagnetic latitudes, namely (Espinosa et al., 2019; Liu et al., 2018; C. Barbosa, Alves, et al., 2015; Barbosa, Hartmann, & Pinheiro, 2015; Gaunt, 2014; Trivedi et al., 2007). In the case of Mexico (~25°N–−40°N), the National Space Weather Laboratory (LANCE) and the National Electricity Operator (Comisión Federal de Electricidad (CFE)), have been collaborating to study the power grid's vulnerability to GIC during space weather events since 2018.

Caraballo et al. (2020) presented the first model for estimating GIC in a subnetwork of 400 kV substations of the Mexican power grid. This initial approach aimed to identify the grid's behavior during geomagnetic storms for the most vulnerable substations. This first model assumed uniform ground conductivity and a uniform geomagnetic disturbance (δB) throughout Mexico. Using the first model, we obtained rough estimates of GIC intensities during severe geomagnetic storms, suggesting that GIC events might be significant in the Mexican power grid. We used these results to plan the installation of GIC sensors at some 400 kV substations.

In this paper, we present the results of an improved numerical model and compare its estimates with 4 months of GIC measurements collected at a coastal substation in 2021 (including five geomagnetic storms). The structure of the paper is as follows: Section 2 describes the Mexican power grid, the new 1D piecewise model, the calculation methodology, and the treatment of measurements. Section 3 discusses the model results and observations. Section 4 compares the new model output with the observed data and the former model's output. Finally, Section 5 presents the conclusions.

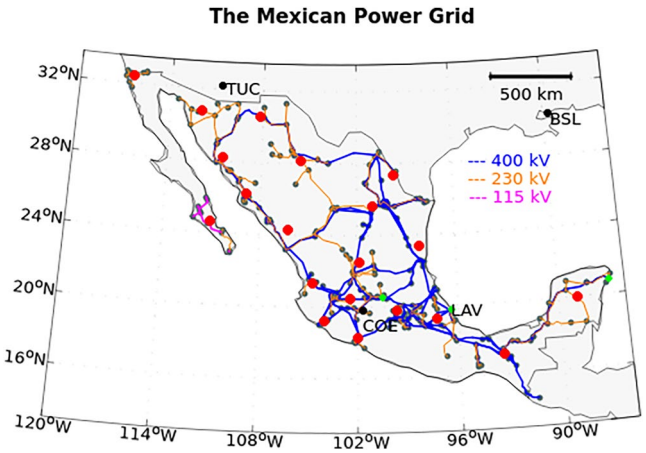


Figure 1. Mexican power grid. The blue, orange, and magenta lines indicate 400, 230, and 115 kV branches. Gray dots are substations (with Laguna Verde marked as LAV), the green ones indicate the location of geomagnetically induced currents sensors. Red points stand for the clustering centers, and black dots pinpoint the Tucson (TUC), Stennis Space Center (BSL), and Coeneo (COE) geomagnetic stations, respectively.

2. Numerical Modeling of GIC in the Mexican Power Grid

2.1. The Mexican Power Grid

The Mexican power grid comprises several subnetworks operating at 400, 230, and 115 kV, plus other lower-voltage subnetworks. In addition, the grid has various connections with the United States, Guatemala, and Belize. In the first model, we only considered 115 substations of the 400 kV subnetwork; see Figure 1 in Caraballo et al. (2020). The new model, however, comprises 246 substations, including 125 substations at 400 kV, 111 substations at 230 kV, and 10 substations at 115 kV (Figure 1). This schema includes 99% of the 400 kV and about 45% of the 230 kV subnetworks. The 115 kV branches correspond to the isolated system in Baja California Sur. We considered those 230 kV branches intersecting the 400 kV system and those connecting remote substations (i.e., with $d \geq 80$ km). We neglected short branches at 230 kV, including the power rings encircling the large urban areas, because of the proximity between substations. We also neglected the lower voltage subnetworks because they have higher resistances and connect to secondary windings of power transformers. We represented these low voltage networks through Thèvenin equivalents connected to some substations (Boteler et al., 2013).

Figure 1 shows several lines parallel to the coastlines and others connecting substations between the Gulf and the Pacific Coasts. This configuration of the Mexican power grid might promote the development of significant GIC during geomagnetic storms. The coast effect can affect lines connecting both oceanic coasts, while the induction along the wires can affect longer lines (Liu et al., 2018). Additionally, most of the substations of the 400/230 kV system have autotransformer banks, which can suffer GIC effects (IEEE, 2015a; Molinski, 2002).

We used the Lehtinen-Pirjola method to calculate GIC values by considering multiple operation voltages and the geomagnetic disturbance calculated throughout the entire power grid (Lehtinen and Pirjola (1985); see Appendix section). Network parameters were extracted from CFE's documentation, while we obtained the grid topology from public domain GIS data available at the OpenStreetMap website (www.openstreetmap.org). First, we set the conductor resistances to 0.051 Ω/km and 0.062 Ω/km for the 400/230 and 115 kV subsystems, respectively. These figures come from the CFE NRF-017-CFE-2008 standard for ACSR 1113 and ACSR 900 conductors (https://lapem.cfe.gob.mx/normas/pdfs/c/E0000-30.pdf). Each power line's phase is composed of a conductor array. Since the GIC are quasi-DC currents ($0.1 - 10^{-4}$ Hz), conductor impedance reduces to their ohmic resistance. We combined all parallel phases and transformer windings into one equivalent line by dividing phase and transformer resistances by 3, as in D. H. Boteler and Pirjola (2014). However, this assumption of balanced GIC through all phases may not be valid under distortion when we extend the network model to the transformer level (Divett et al., 2018; Jankee et al., 2020). In our case, we keep this assumption because we focused on substation modeling. Multiple connections between substations through different pathways were reduced to a unique connection with a series-parallel equivalent set of sources and resistances to account for the induction occurring at each branch. Since we installed the sensors at the common neutral of the transformer banks, we estimate the total GIC intensity at each substation.

Substation hardware parameters are restricted information and thus difficult to access. Following Horton et al. (2012), we set the autotransformer winding resistances to 0.04 Ω/phase and 0.06 Ω per phase for primary and common windings. Likewise, we set the YY transformers' primary and secondary winding resistances to 0.2 Ω/phase and 0.1 Ω, respectively. These values are reasonable estimates according to the operational voltages, the hardware employed, and the values used in previous studies. We assign 0.5 Ω/phase to the grounding resistances, a value within the range recommended for 400–500 kV power substations (IEEE, 2015b). Finally, to include both voltage levels in the model, we included separate neutral points as nodes according to D. H. Boteler and Pirjola (2017). In this case, we considered only the most relevant 230 kV lines intersecting the 400 kV subnetwork.

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2.2. On the Geomagnetic Disturbance

At low-to-mid latitudes, the main sources of the geomagnetic disturbance are perturbations of magneto-ionospheric origin involving the magnetopause current, the ring currents, the equatorial electrojet, and the magnetotail currents. For simplicity, let us consider the geomagnetic disturbance as a plane EM wave propagating vertically into the ground (*i.e.*, *plane wave method*). Assuming a standard geoelectromagnetic coordinate system, where the *xy*-plane represents Earth's surface and the *z*-axis points downwards (Cagniard, 1953; Simpson & Bahr, 2005).

Assuming an in-depth stratified medium, we can neglect the displacement currents for typical frequencies and conductivities at the Earth's surface. Thus, for $\mathbf{E},\mathbf{B} \propto \exp(j\omega t)$ we get:

$$\nabla \times \mathbf{E} = -j\omega\mathbf{B} \tag{1a}$$

$$\nabla \times \mathbf{B} = \mu_0\sigma\mathbf{E}, \tag{1b}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H.m^{-1} is the magnetic permeability constant and $\omega = 2\pi f$, with f the wave frequency.

In a medium of uniform conductivity, there is no charge accumulation; hence $\nabla \cdot \mathbf{E} = 0$. Thus, applying the curl to Equation 1a and using Equation 1b, we get a diffusion equation for the **E** field.

$$\nabla^2\mathbf{E} = k^2\mathbf{E}, \tag{2}$$

and other analogous for the **B** field, where k represent the propagation coefficient given by

$$k = \sqrt{j\omega\mu_0\sigma}. \tag{3}$$

Under the plane wave assumption, x, y lateral variations are much less significant than in the z direction. Hence, Equation 2 reduces to

$$\frac{\partial^2 E}{\partial z^2} = k^2 E. \tag{4}$$

The "transfer function" $K(\omega) = E(\omega)/B(\omega)$ relates both **E** and **B** fields. Analogously, we can define $Z(\omega) = \mu_0 K(\omega)$ as the "surface impedance." We will express all the following relations in terms of $Z(\omega)$.

The surface impedance at the top of a uniform half-space of conductivity σ is given by

$$Z(\omega) = \sqrt{\frac{j\mu_0\omega}{\sigma}}. \tag{5}$$

In an in-depth stratified medium, the surface impedance $Z(\omega)$ at the top of the nth layer of thickness d_n is given by the recursive relation:

$$Z_n(\omega) = \eta_n\left(\frac{1 - r_n e^{-2h_n z}}{1 + r_n e^{-2h_n z}}\right). \tag{6}$$

Where η_n is the characteristic function of the nth layer defined by

$$\eta_n = \sqrt{\frac{j\omega\mu_0}{\sigma_n}}. \tag{7}$$

and r_n represents the reflection coefficient at the bottom of the nth layer, which is given by

$$r_n = \frac{\eta_n - Z_{n+1}}{\eta_n + Z_{n+1}}. \tag{8}$$

We start iterating from the deepest layer of conductivity σ_N, which is considered a uniform half-space with a surface impedance given by Equation 5 up to the surface. In the last step, we get $Z(\omega)$, the surface impedance of the whole stratified medium.

During this study, Mexico had only one magnetic station, located at Coeneo (see Figure 1). GIC modeling requires calculating the geomagnetic disturbance δB at each point throughout the Mexican territory. In this case, we

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

interpolated δB by using Spherical Elementary Current Systems (SECS) (Amm, 1997). This robust non-spectral technique allows us to estimate δB even with data from a few distant observatories (Caraballo et al., 2013; Heyns et al., 2022). The SECS algorithm simulates the external source of δB by placing a system of divergence-free current poles at the mean height of the E layer (≈110 km). We represented the actual magneto-ionospheric currents with mathematical equivalents that produce the same magnetic disturbance. Once the divergence-free currents are solved, we calculate δB at a given location by locally inverting such currents. In addition, to include the effects of the induced EM field at the surface, we can also place telluric poles at a complex depth p defined as

$$p = \frac{Z(\omega)}{j\omega\mu_0}, \tag{9}$$

where $Z(\omega)$ is the ground impedance and ω and μ_0 are frequency and vacuum permeability, respectively (D. H. Boteler & Pirjola, 2002; D. Boteler & Pirjola, 1998). Ground conductivity plays an important role here, as seen in the next section. This separation between ionospheric and telluric contributions was performed in the past by Pulkkinen et al. (2003). To calculate δB, we used input data from the magnetic observatories in Tucson, Arizona, in the United States (39.30°N, 42.40°W) and Stennis Space Center (39.26°N, 18.97°W), as well as the LANCE magnetic station in Coeneo in central Mexico (27.94°N, 30.98°W), all places given in geomagnetic coordinates at epoch 2021.0. We placed a grid of 143 divergence-free current poles and their corresponding telluric images at depth p distributed in a grid of (13×11) points with a separation of 2° in latitude and 4° in longitude, respectively. The grid of div-free current poles covers the entire area shown in Figure 1.

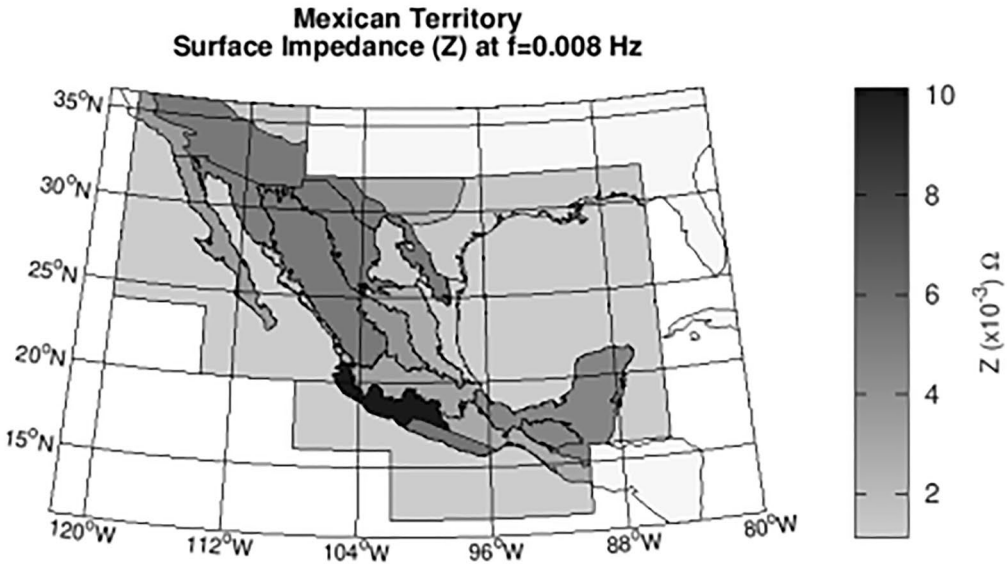
The SECS algorithm estimates δB at high latitudes, where field-aligned currents (FAC) are almost perpendicular to the Earth's surface. At low latitudes, it may be necessary to account for the inclination of the FAC currents among other current sources. Hence, we introduced two corrections to account for slant FAC and a few observation points. In the first case, we approximate the slant FAC by an inclined current line in the meridional plane. This approximation introduces a slight asymmetry in the E-W geomagnetic field due to the inclusion of a curl-free contribution. Fortunately, this is a small effect unless we locate it over the magnetic equator. For the second correction, we introduced three virtual observation points. We calculated these virtual observations using a latitudinal weighted average of the actual observed data. These corrections help to keep the output within a range characteristic of such latitudes. In the Figure S4 in Supporting Information S1 shows the interpolation results of δB for COENEO with data from magnetic observatories located in Tucson and the Stennis Space Center, respectively. In the Figure S5 in Supporting Information S1 shows the location of all the real and virtual observation points. Regarding interpolation, we appreciate a good correlation between the observed and interpolated geomagnetic disturbances in each horizontal component.

The substations are unevenly distributed throughout the Mexican territory; Figure 1 shows that they are concentrated around large urban areas. The mean distance between nodes in the 400 kV subnetwork is ≈105 km, whereas it is about 15 km for the 230 kV one. To avoid wasting time interpolating δB for closer locations. We used a K-nearest neighbor algorithm (KNN) to determine clustering centers (i.e., centroids) and where to interpolate the δB within each cluster. Weighing computational efficiency and accuracy, we partitioned the whole set of nodes (i.e., substations) into 20 clusters. With this number of centroids, nodes inside each cluster are within a radius of less than 120 km between each other, which is in the range of spatial coherence for low latitude geomagnetic data (Ngwira et al., 2008). KNN is a well-known machine learning algorithm for classification (Altman, 1992). Therefore, instead of interpolating δB at each point along the grid, we interpolated at the centroids (the red dots in Figure 1). Thus, we assumed the same δB for points within each cluster, which significantly reduces the computing time.

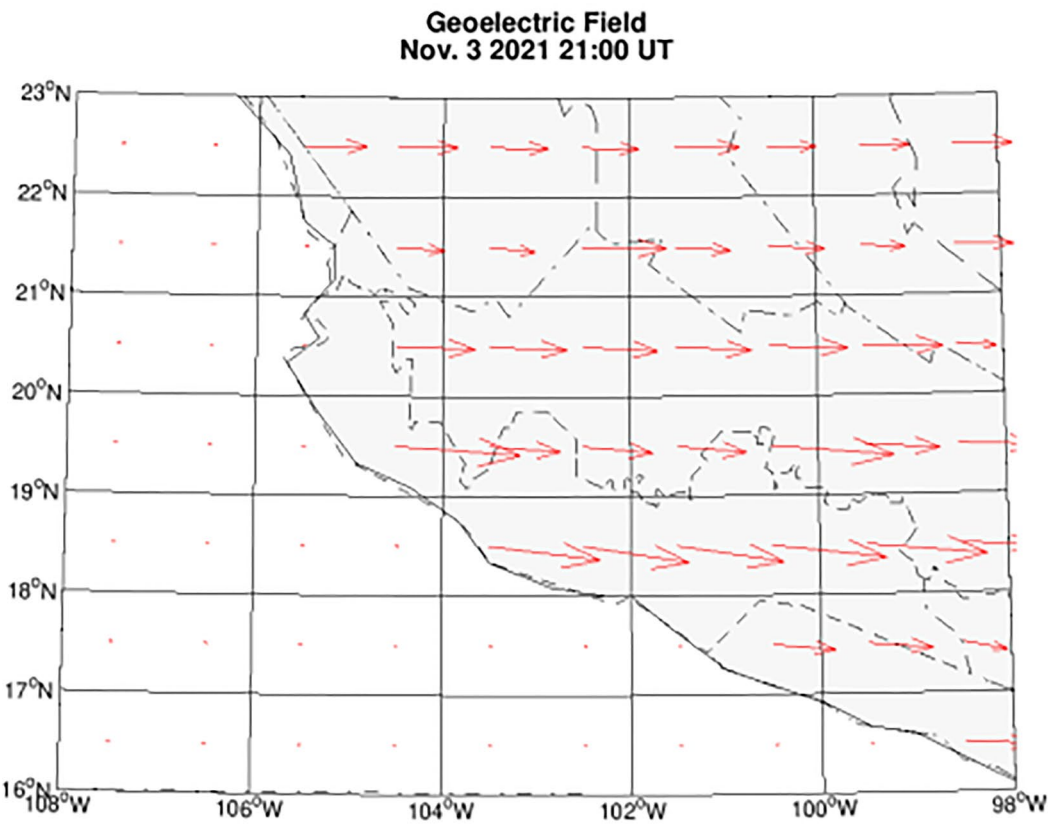
2.3. The Ground Conductivity Model

GIC power spectra for low-to-mid latitudes suggest that these regions typically have dominant frequencies below 0.005 Hz (Oyedokun et al., 2020). GIC currents can penetrate ≥225 km into the ground at these frequencies for an average ground resistivity of 1,000 Ω m. Hence, the GIC can propagate below the crustal-upper mantle boundary.

We based the new conductivity model on data from broadband magnetotelluric (MT) surveys distributed in central and southern Mexico (Corbo-Camargo et al., 2013; Delgado-Rodríguez et al., 2001; Jording et al., 2000; Sedlock et al., 1993) and the southern United States (Fernberg, 2012; Kelbert et al., 2019). As shown in Figure 2a



(a)



(b)

Figure 2. (a) Surface impedance for an electromagnetic perturbation with a period $T = 120$ s (0.008 Hz), according to the ground conductivity model proposed for the Mexican territory and surroundings. Each block corresponds to a physiographic province with a vertical conductivity structure. (b) Geoelectric field during the Nov. 03 geomagnetic storm onset (21:00 UT) along the shoreline of Oaxaca physiographic province and their surroundings. We can appreciate significant changes in the E-field vector at the boundaries with the ocean. The longest and shortest arrows correspond to 30 mV/km and 0.1 mV/km, respectively.

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

we modeled the Mexican lithosphere and upper mantle conductivity through a mosaic of 15 stratified blocks, such as the EURHOM model (Ádám et al., 2012). Each block (i.e., physiographic province) has an in-depth 1D conductivity profile. Text S3 in the Supporting Information S1 presents a detailed description of the 1D piecewise model used. We obtained the original physiographic profiles from Mexico's Instituto Nacional de Estadística (INEGI) and the US Geological Survey. In some cases, the original shape files were modified to ease the concordance with all these studies.

We modeled the surrounding oceanic areas as a half-space of uniform conductivity $\sigma = 4$ S/m. This is a relatively simple approach, considering the strong attenuation of the electromagnetic waves with the depth of the oceanic medium.

Figure 2a shows the distribution of the conductivity blocks and their surface impedance for a 0.008 Hz EM perturbation ($T = 120$ s). Higher surface impedance (Z) values occurred in the Yucatán Peninsula, the Oaxaca coast, and the northwestern side of Mexico.

Lateral conductivity discontinuities along coastal areas may contribute to the increase of the local geoelectric field because of charge accumulation at these boundaries (Jones & Price, 1971; Liu et al., 2019). This effect might enhance the induction effects on nearby power lines. Unfortunately, this simple 1D piecewise model cannot account for E and H polarization effects occurring at the shorelines or at the boundaries between different geological provinces. Hence, we can expect some underestimation in the calculated E-field for points close to these areas. The Figure 2b shows the E-field along the boundaries of the Oaxaca province at a given instant during the onset of a geomagnetic storm. We can see the most prominent changes in $\mathbf{E}$ at the boundaries with the ocean. This can be attributed to the high conductivity of the oceanic provinces. Once the geoelectric field is determined, the voltage difference between grid nodes V_{ij} is found by integrating along the power lines.

3. Results

3.1. On the GIC Measurements

The GIC sensor consists of a Hall effect transducer placed between the transformer's neutral and measures GIC in ±50 A with a precision of ±5%. Additionally, we measured the transformer's temperature through a thermal transducer placed at its hull.

We have GIC measurements for 2021 at the Laguna Verde (LAV) substation near Veracruz; (28.2°N, 25.67°W geomagnetic coordinates, see Figure 1). Here, we focused on data from the LAV substation between August and November 2021 (near the minimum of solar cycle 24).

We used a modified version of the Whitaker-Hayes algorithm to remove spikes and fix small gaps in the raw data (Whitaker & Hayes, 2018). The GIC signal seems to be contaminated with sporadic noise with $T \leq 20$ min. We attribute this issue to external sources, such as nearby electromechanical devices or some electronic noise; this issue is being investigated.

Additionally, a Hodrick-Prescott filter with tuning parameter $\lambda = 1{,}600 \cdot 4^4$ was used to remove diurnal and seasonal trends on the temperature data (Hodrick & Prescott, 1997). Hence, we obtained the hull's residual temperature ΔT, which is a proxy for thermal effects in the transformer. Thermal effects on transformers caused by GIC have been studied by several authors (see Vakhnina et al. (2016) and references therein).

Five geomagnetic storms occurred from August through November 2021. Figure 3 shows the entire period at a glance; black lines in the first-panel mark each event's onset. The first three events were G1 storms, while the last two corresponded to G2 storms. This weak-to-moderate activity is typical around the solar cycle minimum. In all these events, there is an enhancement in the observed and calculated GIC (blue and red curves, middle panel). The top panel shows the estimated geoelectric field components at the LAV substation. There are additional peaks in E_x and E_y that do not appear in the GIC plot. These peaks can be ascribed to small gaps in the magnetic data. The FFT used can manage time series with gaps, but at the cost of introducing a small amount of harmonic distortion. Some peaks are still present in the modeled GIC in an attenuated form; this is a purely numeric effect. The lower panel shows the K_p index as a proxy for global geomagnetic activity.

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

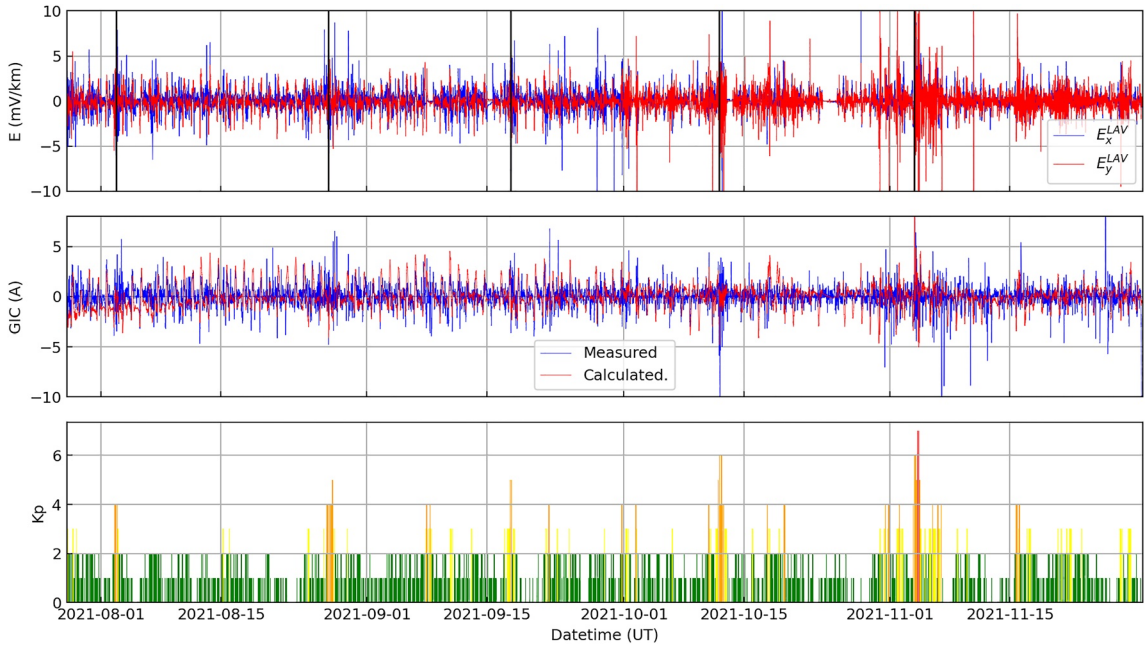


Figure 3. Geoelectric Field and geomagnetically induced currents (GIC) values at the Laguna Verde (LAV) substation (August to November 2021). The upper panel illustrates the calculated northward and eastward components of the geoelectric field (E_x, E_y). Vertical black lines mark the onsets of the five geomagnetic storms during this period. The middle panel shows the calculated (red) and measured (blue) GIC at LAV. The lower panel shows the Kp index as a proxy of global geomagnetic activity. The red, orange, yellow, and green bars indicate $Kp \geq 6$, $Kp = 5$, $Kp = 4$, and $Kp < 4$, respectively.

3.1.1. Geomagnetic Quiet Conditions

Theoretically, the GIC current through a transformer tends to be null during quiet geomagnetic conditions. There can, however, be a small contribution from other sources, like electrochemical interactions between the soil and the grounding mat. We studied the GIC measured at the LAV substation during quiet geomagnetic periods to determine its lowest level. Thus, we performed a superposed epoch analysis (SEA), choosing 78 quiet days ($Kp \leq 2$) between July and November 2021 with a time window of 1 day centered at noon. Once the daily time series were stacked and centered, each value was averaged along the stacking axis. The SEA analysis helps to discriminate the actual GIC signal from the background noise because the last will tend to average zero. In Figure 4, the red and blue curves represent the median of the distribution of values along the stacking axis for the observed and modeled GIC. The shaded areas in both panels correspond to the interquartile range (IQR) of each data distribution. The IQR depicts 50% of the distribution of values and is a robust measure of their dispersion. We can see that variations in the GIC current are within the range of ±1.0 A. This may be attributed to the diurnal variation. On the other hand, between 18:00 and 5:00 hr (UT), GIC intensities remain close to the instrumental noise level.

As the GIC data contains gaps, metrics like the Pearson correlation do not perform well. On the other hand, the root-mean-square error (RMSE) usually penalizes large deviations over small ones. To address these shortcomings, we argue that a metric aimed at the overall signal aspect could be more appropriate for this case. Thus, we use a *relative mean deviation* (RMD), defined as:

$$\mathrm{RMD} = \left.\frac{\langle |x - \hat{x}| \rangle}{\langle |x| \rangle}\right|_{|x|>x_{th}}, \tag{10}$$

where $x, \hat{x}$, x_{th} are the observed, calculated, and threshold values, respectively. The RMD coefficient is a scale-independent, dimensionless accuracy metric that compares measured and predicted values above a given threshold. The purpose of the threshold is to filter only relevant data. A similar metric has been used in previous studies to assess the goodness of fit in SECS interpolations, (Juusola et al., 2006). The RMD metric is sensitive to discrepancies between actual and predicted values and measures the overall relative error. Therefore, low RMD values indicate a good agreement between observations and predictions.

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

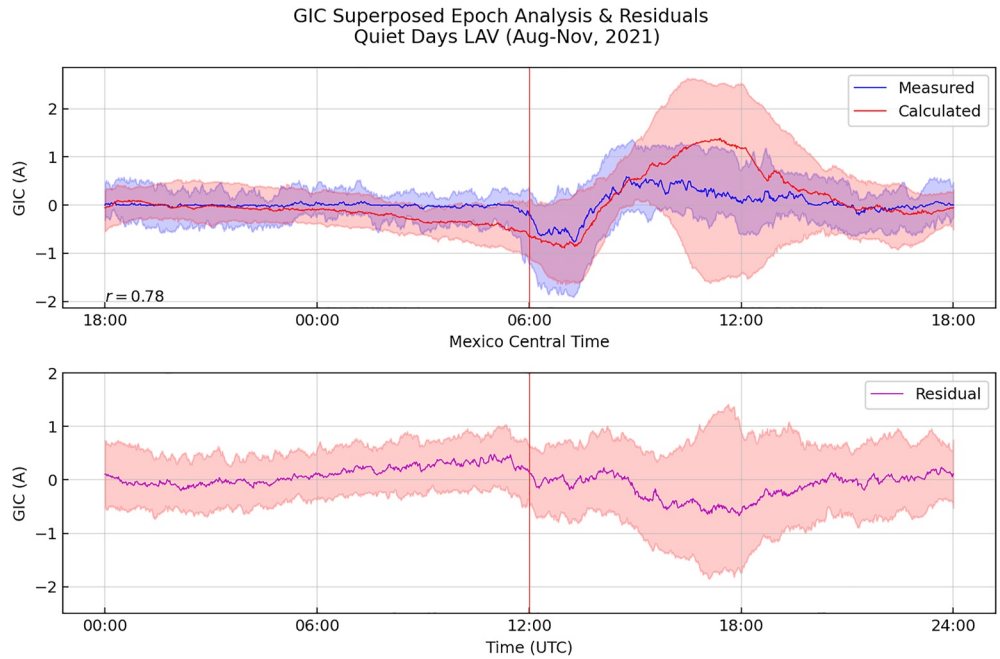


Figure 4. (Upper panel) Superposed Epoch Analysis, for the calculated (red) and observed (blue) geomagnetically induced currents (GIC) values at the LAV substation during quiet days (August–November 2021). In the lower panel, we show the residuals between the observed and calculated GIC with their respective confidence bands. The shaded areas in both panels correspond to the interquartile range (IQR) of the distribution of values. The horizontal axes show both the Central Mexican and UT times, respectively.

The lower panel in Figure 4 shows the differences (residuals) between calculated and observed GIC values. Again, shaded regions represent the interquartile range (IQR) of the residuals. An interesting correlation exists between both series of medians, according to their value $r = 0.87$. This result suggests that the minimum GIC intensity at the LAV substation, during geomagnetically quiet periods, varies within the interval of ± 1 A. This value agrees with observations from other low-latitude countries, and we can consider it the minimum GIC level above the experimental noise. This variation can be attributed to the diurnal variation because it has the same periodicity (C. Barbosa, Alves, et al., 2015; Barbosa, Hartmann, & Pinheiro, 2015; Oyedokun et al., 2020).

3.1.2. Geomagnetic Storm Events

Table 1 shows the geomagnetic storms that occurred in the period, with the strongest one occurring on 4 November (Dst = −115.3 nT). We calculated the RMD within a 3-day window centered on the day of the storm's onset. We set the threshold value $x_{th} = 1.5$ A according to the results in Section 3.1.1.

Table 1
Five Geomagnetic Storms Occurred From August to November 2021

Date	Dst (nT)	Class	GIC_{max} (A)	GIC_{max}^{model} (A)	RMD
2 August	−30.3	G1	3.4	3.0	0.56
28 August	−75.6	G1	6.4	4.0	0.84
17 September	−76.4	G1	4.2	2.8	0.50
12 October	−69.5	G2	11.3	4.8	0.83
4 November	−115.3	G2	12.5	8.8	0.67

Note. Table columns show (a) the minimum Dst index for each event, (b) storm class (NOAA classification), (c) the maximum GIC value detected at the LAV substation, (d) the maximum modeled GIC, (e) RMD, the misfit.

The Figure 5 shows four of the GIC events analyzed (a fifth event is shown in Figure S2 in Supporting Information S1). The upper panels correspond to G2 geomagnetic storms, whereas the lower ones show G1 events. The insets in each panel depict the degree of correlation between both time series and the RMD/RMSE metrics for comparison. We can see that the GIC values at the LAV substation barely surpassed 5 A during the G1 storms. On the contrary, for the G2 storm on 4 November, the observed GIC exceeded 10 A for approximately 25 min. The RMD increases in some events, denoting a significant misfit in values above the threshold. This is because additional peaks not present in the predicted values decrease the overall correlation, (i.e., increasing the RMD). These additional variations can be attributed to other causes not correctly accounted for by the model. The events with lower RMSE values indicate that the calculated GIC does not depart more than 2 Amps on average from the peak value of the observations.

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

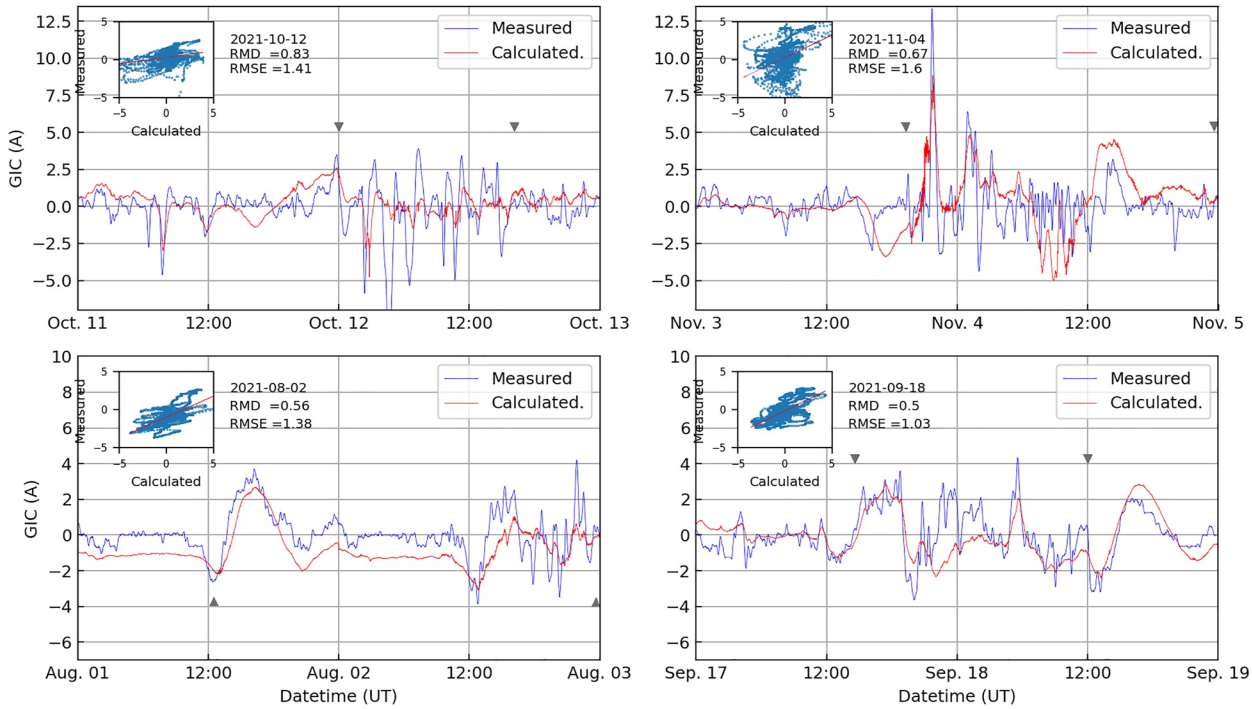


Figure 5. Observed geomagnetically induced currents at LAV during four geomagnetic storms between August and November 2021. The upper panels correspond to G2 geomagnetic storms, and the lower ones correspond to G1 storms. Annotations indicate the date of the storm onset and the relative mean deviation and RMSE metrics, respectively. Black arrows indicate the storm intervals. The scatter plots in the insets show the correlation curve between measured (*y*-axis) and calculated values (*x*-axis).

3.2. Thermal Variations

One of the main concerns for the electric industry is the thermal damage produced in transformers because of overheating events. The purpose of the temperature sensors is to quantify the device's thermal stress. The GIC can cause vibration, saturation in the transformer's core, and reactive power consumption, leading to overheating.

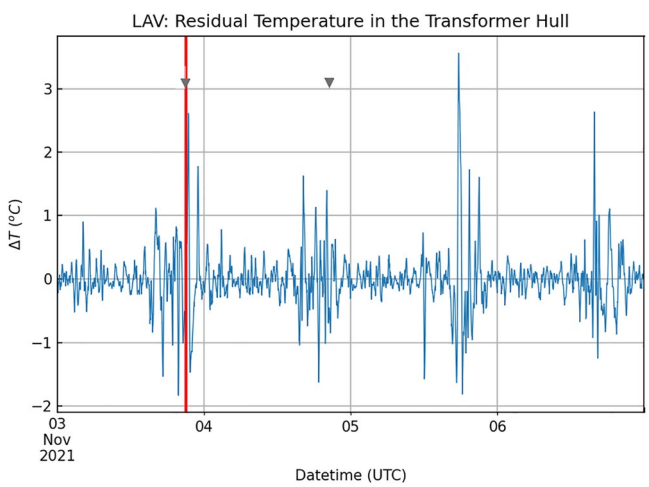


Figure 6. Daily variation of the residual temperature in a 400 kV autotransformer hull at LAV substation. Data correspond to the week of 4 November 2021. Black arrows mark the geomagnetic storm interval, and the red line indicates the storm's onset time near 21:00 hr on 3 November 2021.

Figure 6 presents the residual temperature (ΔT) for a 400 kV transformer at the LAV substation on November 3–7, 2021. We see a cyclic perturbation with $T \approx 12$ hr, which peaks ~18 hr UT (12 hr local time). This can be attributed to the device's working regime. There were no significant variations during the geomagnetic storm on November 4, maybe due to the weak intensity of the geomagnetic perturbations. However, this subject may require further study in combination with careful transformer-level modeling.

4. Discussion

4.1. Power Grid Response to Geomagnetic Storms

We studied the 1D piecewise model performance during geomagnetic quiet and disturbed conditions. The comparison between observations and estimates of GIC (in Table 1) shows that the RMD is within the range of 0.56–0.84 for the events considered. Figures 3 and 5 show differences between observed and calculated peaks. However, the modeled data follow the main trend of the observed signal. We ascribe these differences to noise from external sources at the nearby or local features not correctly accounted for by the model. Regarding the latter, the insufficient number of observational sources used to perform the δB interpolation, inaccuracies in the ground conductivity, and incorrect network information may also significantly impact the results.

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

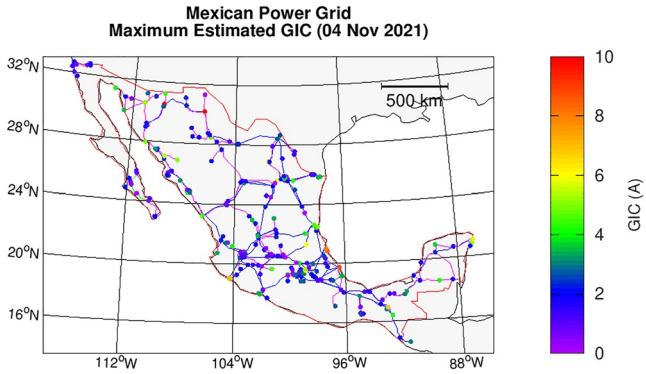


Figure 7. Maximum geomagnetically induced currents (GIC) for each substation of the Mexican power grid on 3–4 November 2021 geomagnetic storm. Some coastal substations registered GIC values ~8 A (reddish dots).

There are some similarities between the piecewise and uniform conductivity models outputs. Regarding the most vulnerable nodes in the 400 kV subnetwork, Figure 7 shows the maximum GIC estimated at each network node (i.e., substation) for the geomagnetic storm occurred on November 4. Although it was a moderate event, the GIC reached peak values ≥8 A for several network nodes (reddish points). Comparing their locations with the ones in Figure 2a and with Figure 3 in Caraballo et al. (2020), the most affected nodes tend to be close to the ends of the grid, over high resistivity terrains, or/and in coastal areas where the conductivity contrast is high. Thus, these results tend to agree with those reported in the literature (e.g., Molinski, 2002; Radasky & Kappenman, 2010 and references therein). Power grids in coastal areas, over high-resistivity terrains, or with specific types of power transformers may suffer GIC effects. Furthermore, Zheng et al. (2014) found that the GIC intensities tend to depend on the characteristic lengths of the entire power grid rather than on specific branch lengths.

4.2. Model Comparisons

Table 2 summarizes the differences between the former uniform conductivity model (Caraballo et al., 2020) and the new 1D piecewise one.

Figure 8 compares the GIC estimates from the two models during the geomagnetic storm on 4 November 2021 for the LAV substation. Although the models' outputs seem to correlate, the uniform conductivity model predicts lower amplitudes and consequently, a worsened fit with respect to the observations. This is evidenced by a larger $RMD_u = 0.78$. Such a feature could be attributed to its scope because it employs a limited number of 400 kV branches and neglects lower voltage subnetworks and lateral variations of the ground conductivity. On the other hand, the improvements in the algorithm and the input data led to capturing the primary trend of the observed GIC at the LAV substation. This result suggests that the 1D piecewise model can produce better GIC estimates for the 400/230 kV Mexican power grid.

5. Conclusions

There is an ongoing effort to understand the vulnerability of the Mexican national electric system concerning severe/extreme geomagnetic storms. Therefore, we presented an improved version of the model for GIC calculation in the 400/230 kV grid. The new model considers multiple voltage levels and transformer types. We also used a 1D piecewise ground conductivity model based on MT data available (in Mexico and its surroundings) to provide a better approach to the electric conductivity structure.

Observed GIC data at the LAV substation between August and November 2021 were compared with their respective estimates from the new model. The SEA analysis suggests that GIC intensities are within the ±1 A range, with an error of ±0.5 A during these periods. Furthermore, the averaged time series for modeled and observed data show a good correlation ($r = 0.87$) during geomagnetically quiet periods. Therefore, such intensities can be considered a reference for the minimum GIC levels at this substation.

The comparison between modeled and observed data also shows an interesting agreement during geomagnetic storms, despite some differences that occur at certain peak intensities. This effect can be attributed to

Table 2
Comparison of the Models for Geomagnetically Induced Currents Calculation in the Mexican Power Grid

Model	Ground model	Voltages (kV)	δB	Nodes	Branches
Uniform	σ = constant	400	Uniform	115	147
Piecewise	stratified 1D	400/230/115	Non-uniform	246	338

Note. Column names refer to model type, ground conductivity structure, operational voltages, geomagnetic disturbance type, and the number of nodes and branches, respectively.

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

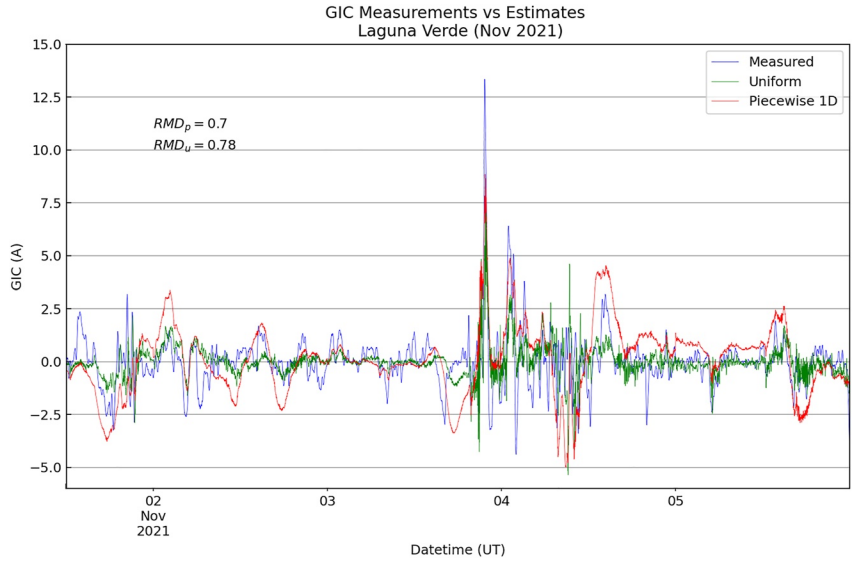


Figure 8. Geomagnetically induced currents estimates at the LAV substation using 1D piecewise (red) and uniform conductivity (green) models for the geomagnetic storm on 4 November 2021. The RMD_p and RMD_u metrics quantify the models' agreement with the observed data (blue).

shortcomings in the input parameters and/or the interpolation of the geomagnetic disturbances, which misses local contributions. However, the piecewise model produces better GIC estimates at the LAV substation than a former uniform conductivity model, which clearly underestimates the observations. Such differences can be attributed to limitations in the scope and resolution of the former model, which only used 115 nodes of the 400 kV grid and assumed uniform ground conductivity.

Observations at the LAV substation show GIC peaks ≥10 A, even during weak geomagnetic perturbations near the solar cycle minimum. This result suggests that space weather phenomena can affect the Mexican power grid regardless of its low geomagnetic latitude. Furthermore, estimates of the 1D piecewise model show that other substations were also affected. These substations are located mostly along coastal areas, in resistive provinces, or close to the ends of the grid.

Appendix A: The Lehtinen-Pirjola Method

The Lehtinen-Pirjola method used for GIC calculation is an improvement in the nodal admittance method (D. H. Boteler & Pirjola, 2017). This method considers the whole grid as a set of n grounded nodes interconnected by admittances y_{ij} and ideal current sources $\mathbf{J_{ij}}$ in parallel (Figure A1).

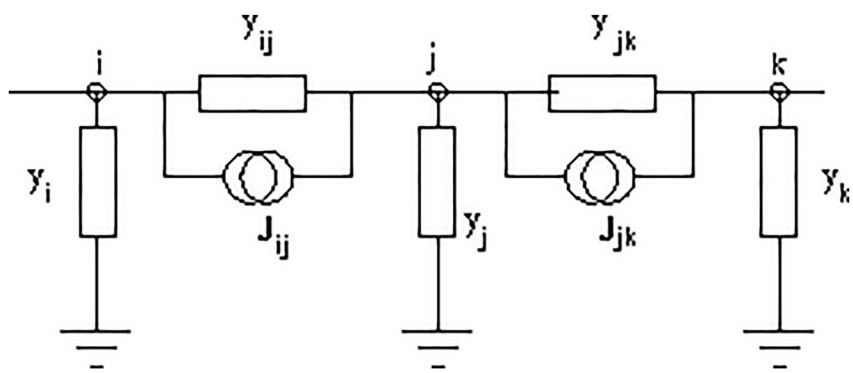


Figure A1. The power grid is represented as a mesh of parallel admittances and ideal current sources. Adapted from Boteler and Pirjola (2017).

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$y_{ij} = 1/R_{ij}$ represents the admittance of the line between i and j nodes in terms of R_{ij} (the resistance of the line between them).

$y_i = 1/r_i$ is the i-node grounding admittance in terms of its grounding resistance r_i.

Defining $\mathbb{Y}, \mathbb{Z}$ as the admittance and earthing impedance matrices, respectively, the GIC current through each grid node is given by

$$GIC = (\mathbb{I} - \mathbb{Y}\mathbb{Z})^{-1}\mathbb{J}, \tag{A1}$$

where $\mathbb{I}$ is the $n \times n$ identity matrix and $\mathbb{Y}$, the admittance matrix given by

$$Y_{ij} = \begin{cases} -1/R_{ij}, & i \neq j \\ \sum_{i \neq j} 1/R_{ij}, & i = j\,. \end{cases} \tag{A2}$$

If the grid nodes are located farther than 1 km apart ($d > 1$ km), the earthing matrix $\mathbb{Z}$ assumes a diagonal form: $Z_{ij} = \mathrm{diag}(r_i, i = 1 \ldots n)$. Finally, the last term on the right of Equation A1, $\mathbb{J} = -V_{ij}/R_{ij}$, represents the earth-free currents given in terms of V_{ij}, which is the induced voltage between the nodes i and j. By solving Equation A1, we obtain the GIC flowing at each grounding point of the grid.

Data Availability Statement

The authors comply with AGU's data policy regarding FAIR Data standards. Readers can access the GNU Octave scripts used in this paper at the OSF general repository (Caraballo, 2023), folder "GIC_assess_v2:SW Journal Article 2022/scripts." GNU Octave it is free software under the terms of the GNU General Public License (Eaton et al., 2020). Coeneo data was obtained from LANCE (http://www.lance.unam.mx/) (Caraballo, 2023), folder "GIC_assess_v2:SW Journal Article 2022/datasets." The results presented in this paper rely on data collected at Tucson and Stennis Space Center magnetic observatories. We thank the USGS for supporting its operation and INTERMAGNET for promoting high standards of magnetic observatory practice (www.intermagnet.org). The SEA analysis was performed with the SpacePy Python package (Morley, S. K. et al., 2011).

Acknowledgments
The authors acknowledge the Comisión Federal de Electricidad (CFE) for their support. The project was partially funded by PAPIIT-IT102720, CONACYT-AEM-292684, and CONACYT-LN 315829. We also thank INTERMAGNET and LANCE for providing the geomagnetic data. Pablo Villanueva and Ernesto Andrade operate the LANCE geomagnetic station.

References

Ádám, A., Pracser, E., & Wesztergom, V. (2012). Estimation of the electric resistivity distribution (EURHOM) in the European lithosphere in the frame of the EURISGIC WP2 project. *Acta Geodaetica et Geophysica Hungarica*, *47*(4), 377–387. https://doi.org/10.1556/AGeod.47.2012.4.1

Altman, N. S. (1992). An introduction to kernel and nearest-neighbor nonparametric regression. *The American Statistician*, *46*(3), 175–185. https://doi.org/10.1080/00031305.1992.10475879

Amm, O. (1997). Ionospheric elementary current systems in spherical coordinates and their application. *Journal of Geomagnetism and Geoelectricity*, *49*(7), 947–955. https://doi.org/10.5636/jgg.49.947

Barbosa, C., Alves, L., Caraballo, R., Hartmann, G. A., Papa, A. R. R., & Pirjola, R. J. (2015). Analysis of geomagnetically induced currents at a low-latitude region over the solar cycles 23 and 24: Comparison between measurements and calculations. *Journal of Space Weather and Space Climate*, *5*, A35. https://doi.org/10.1051/swsc/2015036

Barbosa, C., Hartmann, G., & Pinheiro, K. (2015). Numerical modeling of geomagnetically induced currents in a Brazilian transmission line. *Advances in Space Research*, *55*(4), 1168–1179. https://doi.org/10.1016/j.asr.2014.11.008

Boteler, D., & Pirjola, R. (1998). The complex-image method for calculating the magnetic and electric fields at the surface of the Earth by the auroral electrojet. *Geophysical Journal International*, *132*(1), 31–40. https://doi.org/10.1046/j.1365-246x.1998.00388.x

Boteler, D. H., Lackey, A. J. C., Marti, L., & Shelemy, S. (2013). Equivalent circuits for modelling geomagnetically induced currents from a neighbouring network. In *2013 IEEE power energy society general meeting* (pp. 1–5). https://doi.org/10.1109/PESMG.2013.6672982

Boteler, D. H., & Pirjola, R. J. (2002). The complex-image method for calculating the magnetic and electric fields produced at the surface of the Earth by the auroral electrojet. *Geophysical Journal International*, *132*(1), 31–40. https://doi.org/10.1046/j.1365-246x.1998.00388.x

Boteler, D. H., & Pirjola, R. J. (2014). Comparison of methods for modelling geomagnetically induced currents. *Annales Geophysicae*, *32*(9), 1177–1187. https://doi.org/10.5194/angeo-32-1177-2014

Boteler, D. H., & Pirjola, R. J. (2017). Modeling geomagnetically induced currents. *Space Weather*, *15*(1), 258–276. https://doi.org/10.1002/2016SW001499

Cagniard, L. (1953). Basic theory of the magnetotelluric method of geophysical prospecting. *Geophysics*, *18*(3), 605–635. https://doi.org/10.1190/1.1437915

Caraballo, R. (2023). GIC assessment on the Mexican power grid [Dataset]. OSF. https://doi.org/10.17605/OSF.IO/AV9HT

Caraballo, R., Bettucci, L., & Tancredi, G. (2013). Geomagnetically induced currents in the Uruguayan high-voltage power grid. *Geophysical Journal International*, *195*(2), 844–853. https://doi.org/10.1093/gji/ggt293

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Caraballo, R., González-Esparza, J. A., Sergeeva, M., & Pacheco, C. R. (2020). First GIC estimates for the Mexican power grid. *Space Weather*, *18*(2). https://doi.org/10.1029/2019SW002260

Corbo-Camargo, F., Arzate-Flores, J., Alvarez-Béjar, R., Aranda-Gómez, J., & Vsevolod, Y. (2013). Subduction of the Rivera plate beneath the Jalisco block as imaged by magnetotelluric data. *Revista Mexicana de Ciencias Geológicas*, *30*, 268–281. http://www.scielo.org.mx/scielo.php?script=sci_arttext&pid=S1026-87742013000200002&lng=es&tlng=en

Delgado-Rodríguez, O., Campos-Enríquez, O., Urrutia-Fucugauchi, J., & Arzate, J. A. (2001). Occam and Bostick 1-D inversion of magnetotelluric soundings in the Chicxulub impact crater, Yucatán, Mexico. *Geofísica Internacional*, *40*(4), 271–283. https://doi.org/10.22201/igeof.00167169p.2001.40.4.410

Divett, T., Richardson, G. S., Beggan, C. D., Rodger, C. J., Boteler, D. H., Ingham, M., et al. (2018). Transformer-level modeling of geomagnetically induced currents in New Zealand's South Island. *Space Weather*, *16*(6), 718–735. https://doi.org/10.1029/2018SW001814

Eaton, J. W., Bateman, D., Hauberg, S., & Wehbring, R. (2020). GNU Octave version 6.1.0 manual: A high-level interactive language for numerical computations [Software] [Computer software manual]. GNU. Retrieved from https://www.gnu.org/software/octave/doc/v6.1.0/

Espinosa, K. V., Padilha, A. L., & Alves, L. R. (2019). Effects of ionospheric conductivity and ground conductance on geomagnetically induced currents during geomagnetic storms: Case studies at low-latitude and equatorial regions. *Space Weather*, *17*(2), 252–268. https://doi.org/10.1029/2018SW002094

Fernberg, P. A. (2012). *One-dimensional earth resistivity models for selected areas of continental United States & Alaska (Tech. Rep.)*. EPRI. Retrieved from https://www.epri.com/#/pages/product/1026430/

Gaunt, C. T. (2014). Reducing uncertainty—Responses for electricity utilities to severe solar storms. *Journal of Space Weather and Space Climate*, *4*(27), A1. https://doi.org/10.1051/swsc/2013058

Heyns, M., Lotz, S., Cilliers, P., & Gaunt, C. (2022). Adaptations to a geomagnetic field interpolation method in southern Africa. *Advances in Space Research*, *72*(3), 854–865. https://doi.org/10.1016/j.asr.2022.03.013

Hodrick, R. J., & Prescott, E. C. (1997). Postwar US business cycles: An empirical investigation. *Journal of Money, Credit, and Banking*, *29*(1), 1–16. https://doi.org/10.2307/2953682

Horton, R., Boteler, D., Overbye, T. J., Pirjola, R., & Dugan, R. C. (2012). A test case for the calculation of geomagnetically induced currents. *IEEE Transactions on Power Delivery*, *27*(4), 2368–2373. https://doi.org/10.1109/TPWRD.2012.2206407

IEEE. (2015a). IEEE guide for establishing power transformer capability while under geomagnetic disturbances. IEEE Std C57. (Vol. *163*, pp. 1–50). https://doi.org/10.1109/IEEESTD.2015.7286929

IEEE. (2015b). IEEE guide for safety in AC substation grounding. IEEE Std 80-2013 (Revision of IEEE Std 80-2000/Incorporates IEEE Std 80-2013/Cor 1-2015) (pp. 1–226). https://doi.org/10.1109/IEEESTD.2015.7109078

Jankee, P., Chisepo, H., Adebayo, V., Oyedokun, D., & Gaunt, C. T. (2020). Transformer models and meters in MATLAB and PSCAD for GIC and leakage dc studies. In *2020 international SAUPEC/RobMech/PRASA conference* (pp. 1–6). https://doi.org/10.1109/SAUPEC/RobMech/PRASA48453.2020.9041060

Jones, F. W., & Price, A. T. (1971). Geomagnetic effects of sloping and shelving discontinuities of earth conductivity. *Geophysics*, *36*(1), 58–66. https://doi.org/10.1190/1.1440164

Jording, A., Ferrari, L., Arzate, J., & Jödicke, H. (2000). Crustal variations and terrane boundaries in southern Mexico as imaged by magnetotelluric transfer functions. *Tectonophysics*, *327*(1), 1–13. https://doi.org/10.1016/S0040-1951(00)00166-9

Juusola, L., Amm, O., & Viljanen, A. (2006). One-dimensional spherical elementary current systems and their use for determining ionospheric currents from satellite measurements. *Earth Planets and Space*, *58*(5), 667–678. https://doi.org/10.1186/bf03351964

Kelbert, A., Bedrosian, P. A., & Murphy, B. S. (2019). The first 3D conductivity model of the contiguous US: Reflections on geologic structure and application to induction hazards. In A. G. Union (Ed.), *Geomagnetically induced currents from the Sun to the power grid* (pp. 127–151). American Geophysical Union. https://doi.org/10.1002/9781119434412.ch8

Lehtinen, M., & Pirjola, R. (1985). Currents produced in earthed conductor networks by geomagnetically-induced electric fields. *Annales Geophysicae*, *3*(4), 479–484.

Liu, C., Ganebo, Y. S., Wang, H., & Li, X. (2018). Geomagnetically induced currents in Ethiopia power grid: Calculation and analysis. *IEEE Access*, *6*, 64649–64658. https://doi.org/10.1109/ACCESS.2018.2877618

Liu, C., Wang, X., Zhang, S., & Xie, C. (2019). Effects of lateral conductivity variations on geomagnetically induced currents: H-Polarization. *IEEE Access*, *7*, 6310–6318. https://doi.org/10.1109/ACCESS.2018.2889462

Molinski, T. S. (2002). Why utilities respect geomagnetically induced currents. *Journal of Atmospheric and Solar-Terrestrial Physics*, *64*(16), 1765–1778. https://doi.org/10.1016/s1364-6826(02)00126-8

Morley, S. K., Koller, J., Welling, D. T., Larsen, B. A., Henderson, M. G., & Niehof, J. T. (2011). Spacepy—A Python-based library of tools for the space sciences [Software]. In *Proceedings of the 9th Python in science conference (SciPy 2010)*. https://doi.org/10.25080/Majora-92bf1922-00c

Ngwira, C. M., Pulkkinen, A., McKinnell, L.-A., & Cilliers, P. J. (2008). Improved modeling of geomagnetically induced currents in the South African power network. *Space Weather*, *6*(11), S11004. https://doi.org/10.1029/2008SW000408

Oyedokun, D. T. O., Heyns, M. J., Cilliers, P. J., & Gaunt, C. T. (2020). Frequency components of geomagnetically induced currents for power system modelling. In *2020 international SAUPEC/RobMech/PRASA conference*. https://doi.org/10.1109/SAUPEC/RobMech/PRASA48453.2020.9041021

Pulkkinen, A., Amm, O., & Viljanen, A. (2003). Ionospheric equivalent current distributions determined with the method of spherical elementary current systems. *Journal of Geophysical Research*, *108*(A2), 1053. https://doi.org/10.1029/2001JA005085

Radasky, W. A., & Kappenman, J. G. (2010). Impacts of geomagnetic storms on EHV and UHV power grids. In *2010 Asia-Pacific international symposium on electromagnetic compatibility* (pp. 695–698). https://doi.org/10.1109/APEMC.2010.5475523

Sedlock, R. L., Ortega-Gutiérrez, F., & Speed, R. C. (1993). Tectonostratigraphic terranes and tectonic evolution of Mexico. In *Tectonostratigraphic terranes and tectonic evolution of Mexico*. Geological Society of America. https://doi.org/10.1130/SPE278-p1

Simpson, F., & Bahr, K. (2005). *Practical magnetotellurics*. Cambridge University Press. https://doi.org/10.1017/CBO9780511614095

Trivedi, N. B., Vitorello, I., Kabata, W., Dutra, S. L. G., Padilha, A. L., Bologna, M. S., et al. (2007). Geomagnetically induced currents in an electric power transmission system at low latitudes in Brazil: A case study. *Space Weather*, *5*(4). https://doi.org/10.1029/2006SW000282

Vakhnina, V. V., Kuznetsov, V. N., & Shapovalov, V. A. (2016). The effect of geomagnetically induced currents on the thermal conditions of power transformers. *Russian Electrical Engineering*, *87*(1), 53–58. https://doi.org/10.3103/S1068371216010119

Whitaker, D. A., & Hayes, K. (2018). A simple algorithm for despiking Raman spectra. *Chemometrics and Intelligent Laboratory Systems*, *179*, 82–84. https://doi.org/10.1016/j.chemolab.2018.06.009

Zheng, K., Boteler, D., Pirjola, R., guang Liu, L., Becker, R., Marti, L., et al. (2014). Effects of system characteristics on geomagnetically induced currents. *IEEE Transactions on Power Delivery*, *29*(2), 890–898. https://doi.org/10.1109/TPWRD.2013.2281191

15427390, 2023, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022SW003202, Wiley Online Library on [17/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License